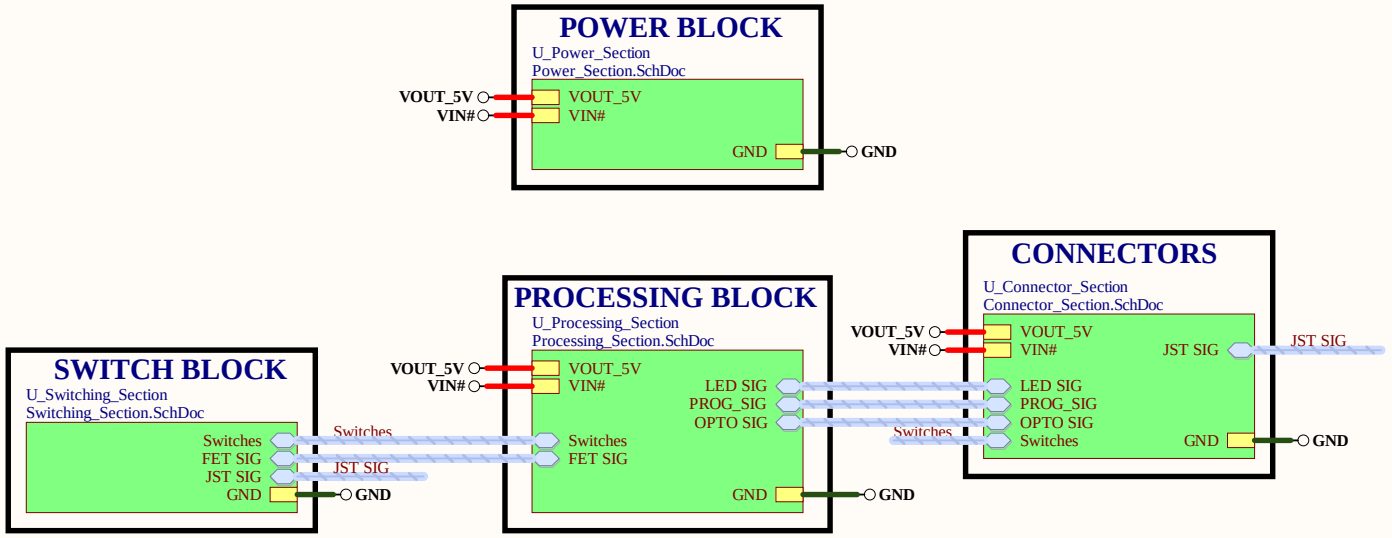
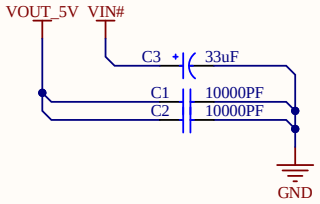
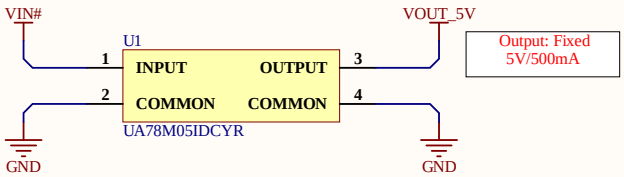

POWER BLOCK
U_Power_Section
Power_Section.SchDoc
VOUT_5V
VIN#
VOUT_5V
VIN#
GND
GND
SWITCH BLOCK
U_Switching_Section
Switching_Section.SchDoc
Switches
FET SIG
JST SIG
GND
GND
PROCESSING BLOCK
U_Processing_Section
Processing_Section.SchDoc
VOUT_5V
VIN#
Switches
FET SIG
LED SIG
PROG_SIG
OPTO SIG
GND
GND
CONNECTORS
U_Connector_Section
Connector_Section.SchDoc
VOUT_5V
VIN#
JST SIG
LED SIG
PROG_SIG
OPTO SIG
Switches
GND
GND
Switches
JST SIG
Switches

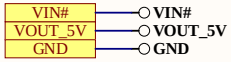
DECOUPLING CAPACITOR

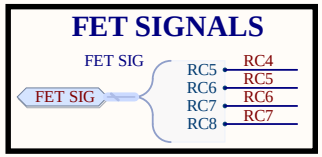


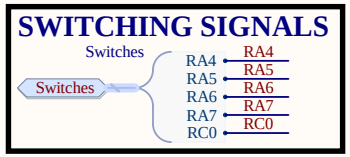
LINEAR REGULATOR

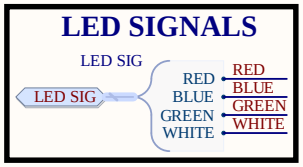


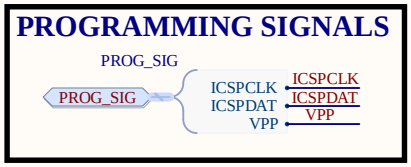
POWER SIGNALS

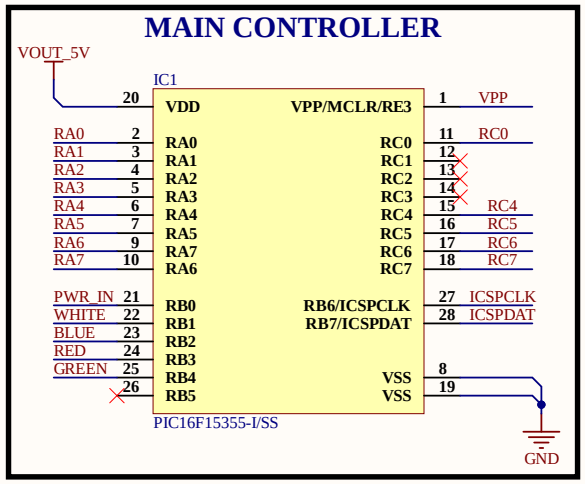


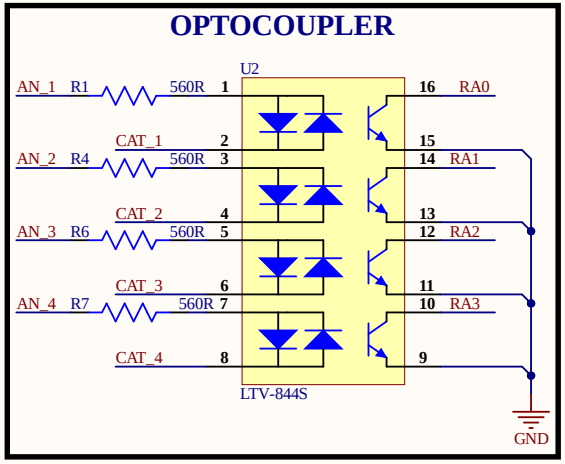

FET SIGNALS
FET SIG
FET SIG
RC5
RC6
RC7
RC8
RC4
RC5
RC6
RC7

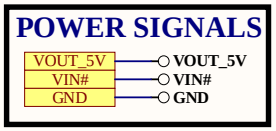

SWITCHING SIGNALS
Switches
Switches
RA4
RA5
RA6
RA7
RC0
RA4
RA5
RA6
RA7
RC0

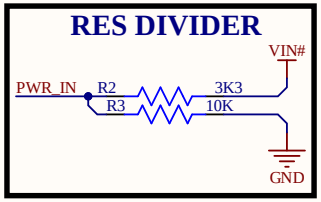

LED SIGNALS
LED SIG
LED SIG
RED
BLUE
GREEN
WHITE
RED
BLUE
GREEN
WHITE

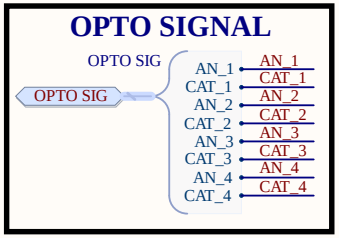

PROGRAMMING SIGNALS
PROG_SIG
PROG_SIG
ICSPCLK
ICSPDAT
VPP
ICSPCLK
ICSPDAT
VPP


MAIN CONTROLLER
VOUT_5V
IC1
20
VDD
VPP/MCLR/RE3
1
VPP
RA0 2
RA1 3
RA2 4
RA3 5
RA4 6
RA5 7
RA6 9
RA7 10
RA0
RA1
RA2
RA3
RA4
RA5
RA6
RA7
RC0
RC1
RC2
RC3
RC4
RC5
RC6
RC7
11 RC0
12
13
14
15 RC4
16 RC5
17 RC6
18 RC7
PWR_IN 21
WHITE 22
BLUE 23
RED 24
GREEN 25
26
RB0
RB1
RB2
RB3
RB4
RB5
RB6/ICSPCLK
RB7/ICSPDAT
27 ICSPCLK
28 ICSPDAT
VSS
VSS
8
19
PIC16F15355-I/SS
GND

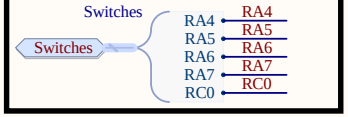

OPTOCOUPLER
U2
AN_1 R1 560R 1
CAT_1 2
AN_2 R4 560R 3
CAT_2 4
AN_3 R6 560R 5
CAT_3 6
AN_4 R7 560R 7
CAT_4 8
16 RA0
15
14 RA1
13
12 RA2
11
10 RA3
9
LTV-844S
GND


POWER SIGNALS
VOUT_5V
VIN#
GND
VOUT_5V
VIN#
GND

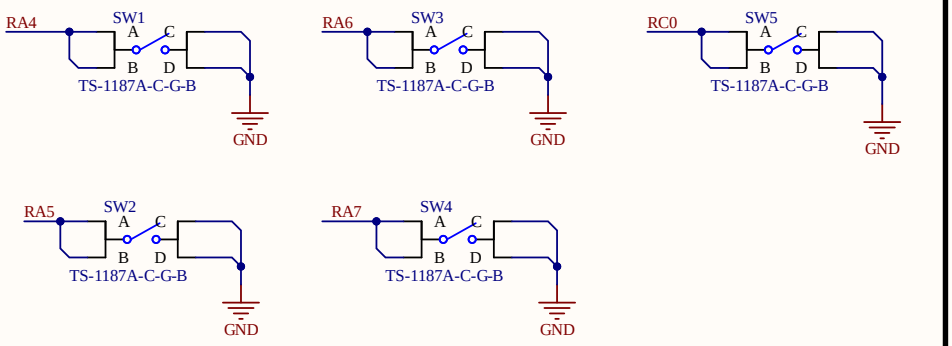

RES DIVIDER
VIN#
PWR_IN
R2
R3
3K3
10K
GND


OPTO SIGNAL
OPTO SIG
OPTO SIG
AN_1
CAT_1
AN_2
CAT_2
AN_3
CAT_3
AN_4
CAT_4
AN_1
CAT_1
AN_2
CAT_2
AN_3
CAT_3
AN_4
CAT_4

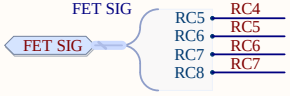
SWITCHING SIGNALS



SW HEADERS



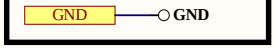
FET SIGNALS



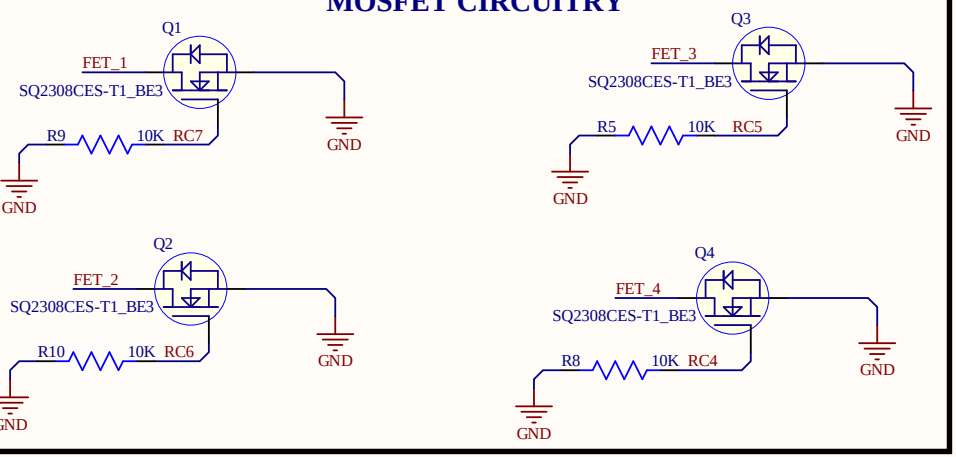
JST SIGNALS



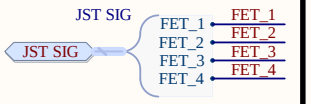
POWER SIGNALS



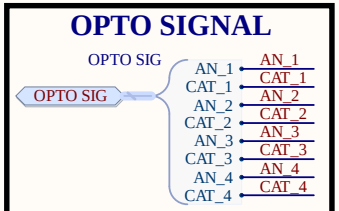
MOSFET CIRCUITRY



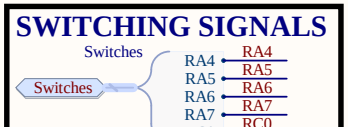
JST SIGNALS



OPTO SIGNAL



SWITCHING SIGNALS



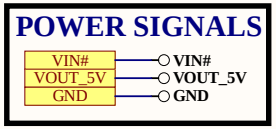
LED SIGNALS



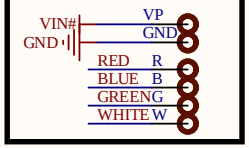
PROGRAMMING SIGNALS



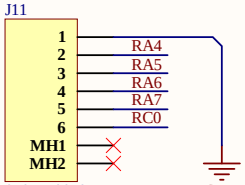
POWER SIGNALS



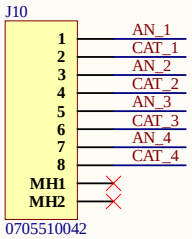
TEST POINTS



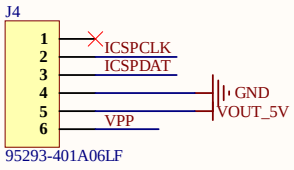
ON-BOARD BUTTON EXTENSION



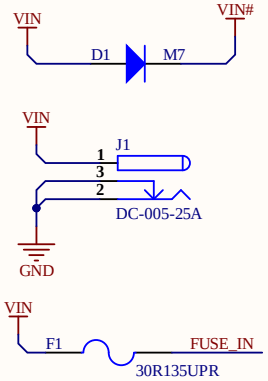
EXTERNAL SENSE SIGNAL



PROGRAMMING HEADER



INPUT POWER



BOARD TO WIRE HEADERS

